# Two Modified Low-light Image Enhancement methods

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## Abstract

Low-light image enhancement (LLIE) aims to improve the visibility and restoring details in low-light images. In this study, two modified methods based on low-light enhancer from paired low-light instance (PairLIE) [1] and LLIE with multi-stage residue quantization and brightness-aware attention (RQ-LLIE) [2], respectively, are proposed. For the first modified method, a new illumination-guided attention block is added, and for the second modified method, an HSV correction net is added.

## Modified Methods

### PairLIE

PairLIE, an unsupervised LLIE network based on the Retinex theory, is trained by paired low-light images. For PairLIE, each image is decomposed into illumination and reflectance components. Assuming that the paired images share the same reflectance components, PairLIE will evaluate the lighting conditions from both images to reconstruct a normal-light image.

L-Net and R-Net are used to evaluate the illumination and reflectance components of the input image, respectively. For the first modified method, two modules, illumination estimator and illumination-guided attention block, are used to assist R-Net to construct the reflectance component.

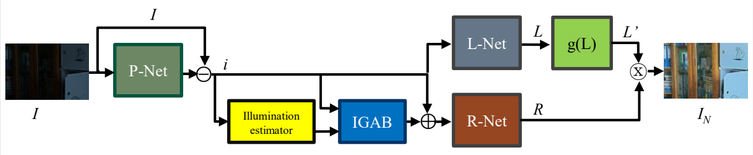



**Fig. 1. Architecture of the modified method with two added modules.**

#### Illumination estimator

Illumination estimator will analyze the light distribution in the image, and enhance the image illumination information.

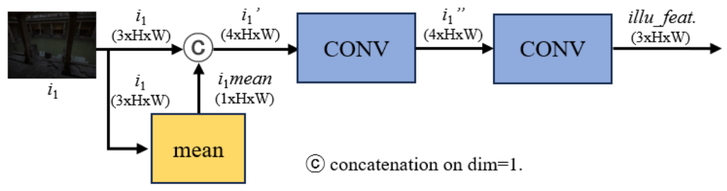



**Fig. 2. Illumination estimator architecture.**

#### Illumination-guided attention block

Illumination-guided attention block (IGAB) is proposed in Retinexformer [3], which is a basic block of illumination-guided transformer (IGT), designed as a corruption restorer.

The light-up process in R-Net may cause under-/over-exposure and color distortion, which are proceeded by IGAB included in R-Net.

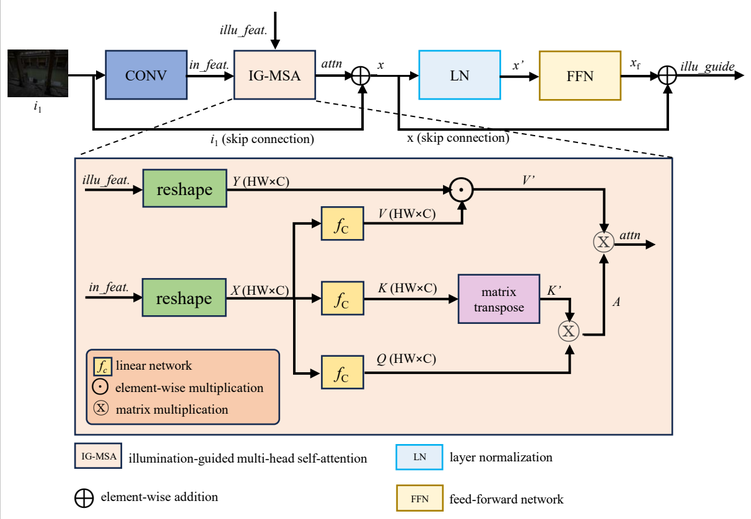



**Fig. 3. Illumination-guided attention block.**

## RQ-LLIE

RQ-LLIE consists of three stages. In stage 1, a codebook is constructed by residual quantization aiming to learn normal light features from the input of normal-light images. In stage 2, a query module is designed to bridge the gap between low-light features and the normal-light codebook. In stage 3, a fusion branch is used to fuse low-light features and normal-light priors from the codebook and the query module.

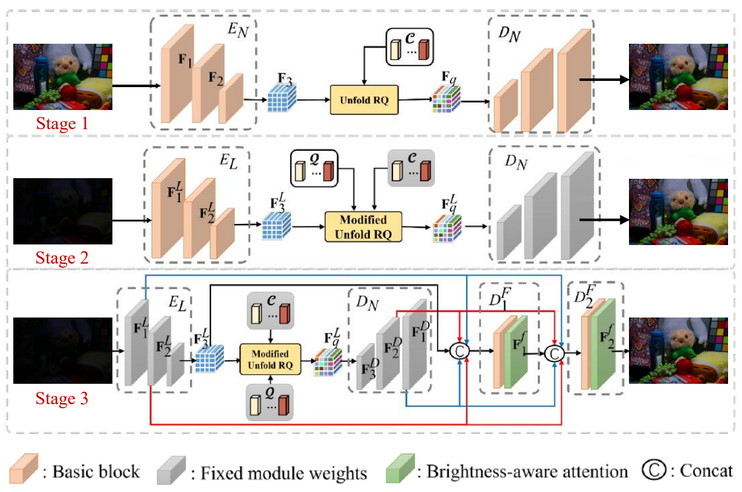



**Fig. 4. RQ-LLIE three-stage framework for low-light image enhancement.**

For the second modified method, RQ-LLIE will followed by an HSV correction net. Each input image will be converted to HSV color space. Then, the saturation and value components will be adjusted by S-Net and V-Net, respectively. Additionally, N-Net is used to convert the image back to RGB color space.

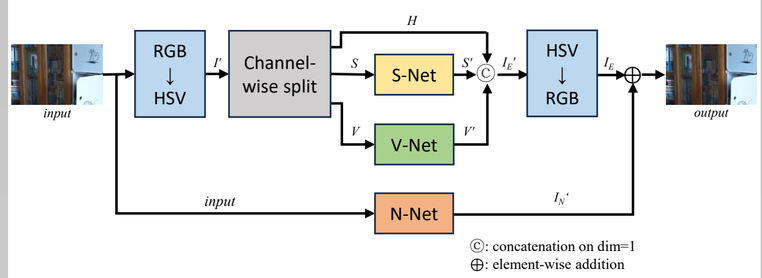



**Fig. 5. Modified method takes the output images of RQ-LLIE for the further evaluation.**

## Experimental Results

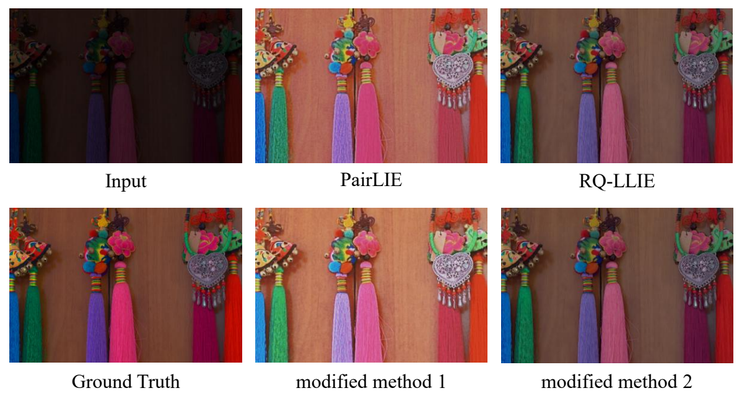



**Fig. 6. Visual comparisons between PairLIE, RQ-LLIE, and two modified methods.**

**Table 1. Performance comparisons between PairLIE, RQ-LLIE, and two modified methods.**

S: Supervised method; U: Unsupervised method

| Methods | Type | LOL | | | |
|---|---|---|---|---|---|
| | | PSNR ↑ | SSIM↑ | LPIPS↓ | Delta E↓ |
| PairLIE | U | 19.51 | 0.736 | 0.248 | 10.80 |
| RQ-LLIE | S | 25.24 | 0.855 | 0.121 | 7.06 |
| Method 1 | U | 19.66 | 0.752 | 0.232 | 11.66 |
| Method 2 | S | 25.55 | 0.861 | 0.116 | 6.91 |

## References

[1] Z. Fu, Y. Yang, X. Tu, Y. Huang, X. Ding and K. -K. Ma, "Learning a Simple Low-Light Image Enhancer from Paired Low-Light Instances," 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Vancouver, BC, Canada, 2023, pp. 22252-22261, 2023.

[2] Y. Liu, T. Huang, W. Dong, F. Wu, X. Li and G. Shi, "Low-Light Image Enhancement with Multi-stage Residue Quantization and Brightness-aware Attention," 2023 IEEE/CVF International Conference on Computer Vision (ICCV), Paris, France, 2023, pp. 12106-12115, 2023.

[3] Y. Cai, H. Bian, J. Lin, H. Wang, R. Timofte and Y. Zhang, "Retinexformer: One-stage Retinex-based Transformer for Low-light Image Enhancement," 2023 IEEE/CVF International Conference on Computer Vision (ICCV), Paris, France, 2023, pp. 12470-12479, 2023.